Digital Data Sovereignty: Towards a Conceptual Framework

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***Abstract*— Sovereignty, which is the dominant concept of political science, is constantly shaping the power relations among modern states externally, and between state and individuals internally, within the context of the territory, the population on the territory, and the activities performed by this population. With the development of digital data-based technologies, the intersection of the physical space with cyberspace, the organic population with the artificial, and the actual activity with the virtual, make the objects that the sovereignty address can and will become increasingly complex. Digital data has become the new enabler of public value. As of uncertainty's new parameter, digital data drives new socio-economic reality. However, the concentration of digital data at non-state actors, reduced need for intermediary institutions and increased non-governmental expertise also lead to transformation of the existing power relations. This paper scrutinizes digitally augmented concept of sovereignty and explores a new conceptual framework to understand the objectives of state sovereignty.**

Keywords—digital data sovereignty, cyberspace, sovereignty, national security, privacy

I. INTRODUCTION

The concept of sovereignty, from the most local political formation to the most global political power relations, is one of the most important concepts that modern politics and modern political science have failed to answer the question of "what" for centuries [1, 2]. On the other hand, modern sovereignty; has been developed in a period where the territory was the land and sea, the population was mostly consisted of settled people and activities carried out by the people in these territories were within sight and physical. While the objects of sovereignty take various new forms in few centuries, the discourse and practice of sovereignty has been transformed, and new meanings are added to the concept according to the existing conditions. For instance, in the air, which has been the new domain of sovereignty for the last 100 years, and finally in outer space, activities such as economic, legal and national security are rapidly becoming more widespread through new technical possibilities. Furthermore, cyberspace that provided by electronic networks has been enabling new and remote social formations and expanding the areas of activity that require sovereign discourse through the technological developments it accelerates. Hence, the state of sovereignty over the complexed different territories and actors has become increasingly controversial.

The validity of the sovereignty claim is consistent with the effectiveness of the techniques owned by the state. Latest economic and social applications of digital technologies have brought to the forefront the national concerns such as cyberspace control, data ownership and authority over autonomous systems. However, the socio-technical basis of the discussions in all three topics has not yet matured. The current debate focuses on ensuring spatial control in a technical sense. On the other hand, the debate on digital data sovereignty as a new technical form of sovereignty in terms of political science and public administration has been centered around the privacy of personal data. Besides, the debate on artificial intelligence-assisted autonomous systems is mostly taking place at the level of possible ethical frameworks. In this context, there are two main discussions. First one is about the effects of digital technologies on sovereign discourse and practices. Second one is about how to realize sovereign practices on contemporary socioeconomics of digital technologies. This study is mostly focus on second one in consideration with first one as a conceptual framework. Although the subject content is essentially referred to as digital data sovereignty, there is a literature with "information sovereignty" [3, 4] and "technological sovereignty" [5].

States primarily aspire to survival, yet it is not the only goal [6]. If this were the case, it is considered that the current states would be shaped according to restrictions. Another important goal of states is to preserve the way of life. They enter competition or cooperation when necessary in order to survive and maintain their way of life. Indeed, the international political system and the balance of power are built on different combinations of competition and cooperation in accordance with the characteristics of these two priorities. Hence, sovereignty claims include survival discourse in tandem with competitive acts.

There is a difference between sovereignty and digital data in terms of tangibility. Sovereignty is, above all, an abstract concept. However, the embodiment attributed to the concept is shaped by the practices that the state directly applies or interferes with. This concrete reality requires authority over the territory and population where the state develops the discourse of sovereignty. On the other hand, digital data is based on concrete realities such as hard disk, database, computer, software, etc. and is a set of "0" and "1" by electrical means. On the other hand, this objective situation makes it possible to isolate technology through information technologies. While the evolving digital data techniques remain within the abstract concept of information technologies, new possibilities can have different consequences on domination under the same concept. This makes it difficult to examine the impact of digital technologies on sovereignty. Where the state practices and the embodiments around digital data intersect, information technologies and the abstractions of sovereignty also intersect. The embodiment of sovereignty can be seen through power relations and through state practices. In order to examine the digital technologies impact on sovereignty, it is necessary to embody the abstraction of the concept of sovereignty and to

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address the concrete practices of the state in the context of sovereignty.

States direct sovereign discourse on territories internally and externally. They instrumentalize these practices as a domestic sovereignty which is for the provision of internal control and authority, interdependence sovereignty for the provision of cross-border crossing control, external sovereignty for the overriding of non-intervention, and international sovereignty for international recognition [7].

In this context, the current characteristics of sovereignty have been detailed in the context of territory, population and activities. Based on this contextual framework, a new perspective to examine the effects of digital data on the concept of sovereignty is discussed.

II. State Sovereignty In Transition

The concept of sovereignty has been significant for modern states and the international system built on this context remain valid as the current balance of political powers. Sovereignty does not naturally accrue to the state. It is a combination of various international and international processes [8]. Today's international political structure is mainly based on the United Nations Convention signed in 1945 after World War II. Article 2 of the convention. In accordance with the principles contained in the Article, all member states have equal sovereignty and member states are obliged to refrain from the use of force to threaten the spatial integrity and political independence of any state [9].

States, in the last 25 years, have faced major events and concepts such as the end of the cold war, the fight against terrorism, the Responsibility for Protection, the increasing number and influence of transnational corporations and the information technology revolution [10]. According to Krasner [11], sovereign state systems have been under unprecedented pressure in the last two centuries under the developments in the phenomenon of globalization and international norms that have changed towards human rights. Globalization challenges internal sovereignty and cross-border crossing in this context, as it makes state control problematic. On the other hand, the transnational norms around human rights make external sovereignty problematic through state authority.

According to Strange [12], financial globalization and the influence of non-state actors on state sovereignty led states to redefine sovereignty according to the new situation. These challenges have led to significant crises in the objective dimensions of sovereignty and in the establishment of authority within the scope of sovereignty. Especially in the post-World War II period, the international understanding based on the objectives set by governments and citizens can be achieved through interstate cooperation has been eroded. a structural understanding of international law and order on behalf of humanity and the world community. Within the scope of this understanding, nation-state and national sovereignty is seen as the new source of resistance and reaction before international new order [13].

The discussions in the literature on the transformation of the objects of sovereignty are generally based on the increasing diversity of the object of sovereignty. The basic element of the concept of sovereignty is the assertion of authority over a territory, a population in this location, and activities carried out by this population. However, when the rhetoric of sovereignty is examined, it is seen that these three elements of sovereignty do not come to the fore simultaneously. For example, "submarine fiber cables" are the object of a sovereign discourse directed towards territory, while "mass surveillance" is the object of a sovereign discourse that observes population and "disclosure of state secret" towards activity.

The discourse of sovereignty seeks to answer the questions "where, who and what?" The answer to these questions is answered differently in line with social development and technological advances. However, the discourse of sovereignty has always targeted these three basic elements, regardless of how the concept is transformed. As a matter of fact, the main objects of the concept of sovereignty consist of a specific territory, population and activity, as represented in Fig 1. In this representation, the objects of sovereignty are represented in three basic dimensions. Accordingly, there is a projection of any sovereignty discourse, as well as space, population and activities.

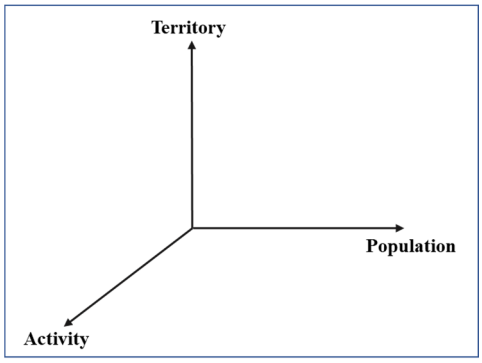


Fig 1. The conceptual objectives of sovereignty

A. Transformation of Territory

Classical definitions of sovereignty were generally defined in the 19th century. It is based on the social formations and the phenomena of land and sea, which mankind can be instrumental in. Considering the technological advances in the 20th and 21st century, the debate on sovereignty over new domains has increased and become the contemporary institutional form. The balance of power of states affected by technological developments at national/international level and the increasing spatial dependence for socio-economic relations are decisive in the transformation of spatial sovereignty. Based on historical development; land, sea, air, space and cyberspace sovereignty can be mentioned. As we move the reality of humanity to new locations and its activities in these spaces diversify, the concept of sovereignty over spaces reproduces itself. State sovereignty historically requires land sovereignty. However;

- Maritime sovereignty in ensuring land sovereignty,
- Air sovereignty in ensuring maritime and land sovereignties,
- Space sovereignty in ensuring air, maritime and land sovereignties,
- Most recently, cyberspace sovereignty, which developed outside of traditional spatializations, has been important for space, air, maritime and land sovereignties.

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Although the claim of the sovereign is defined at absolute range for its initial definition, the current international system is based on inter-state agreements such as the UN Security Council Resolutions and the Geneva Convention. On the other hand, there are exceptions in the agreements due to the conditions of the territories. For example, it is essential that no country claims sovereignty on the high seas. Similarly, all states have common rights on seabeds, Antarctica and the Moon as determined by UN agreements.

While international discussions on territories have reached a certain level of maturity and regulation, as Kittichaisaree [14] pointed out, the phenomenon of cyberspace remains to be regulated internationally. Meanwhile, the relevant provisions of existing international law (UN Convention) generally apply to cyberspace sovereignty. As stated in the Tallinn Manuel 2.0, no state can claim sovereignty directly over cyberspace, while able to address sovereignty over cyberinfrastructures and activities associated with these infrastructures [15].

Countries' cyberspace-related policies are closely related to levels of sophistication and international political objectives in information and communication technologies [14]. For example, the United States, which has global dominance over ICT, emphasizes the free movement of information and fundamental freedoms. On the other hand, Western European countries seek for the protection of privacy and personal data, cyber security in African countries where internet usage is rapidly spreading, and China is legalizing the right to national sovereignty in cyberspace, who has the most internet users. Countries with technologically advanced competence are at risk of monopoly on cyberspace law unless they are restricted by other countries. Indeed, many states are unintentionally consensus or object to what they do not understand, as they are unwittingly indifferent to the results of the relevant technology when they are released and accessible to the public.

B. Transformation of Population

Besides evaluating individuals as a basic social unit through sovereignty over the body, it is possible to determine the scope of a discourse that directs the body to all kinds of organic or inorganic components that make up the individual. In line with advances in digital data-driven biotechnology, today's political space goes beyond the human body's relationship with things, directly to all the parts that make up the human body and the genetic records that make up the living. The field of biotechnology, which has the power to shape the human nature that shapes the diversity of political thought, has the potential to negatively affect the nature of democracy and politics [16].

The main object of population-based sovereignty is individuals, who are legally referred to as real persons. However, in social life, there are some rights and assignments that real persons have and legal entities that have legal equivalents. Legal entities are a social entity, or a community of goods and the work and transactions carried out on behalf of them are carried out by real persons. However, today's sovereignty debate has gone beyond the traditional human institution. Physically and genetically interfered with human, domestic and wild animals, natural life and artificial intelligence issues are also becoming the agenda of political science. The most important distinguishing feature of today's reality is that digital systems in human activities may be reflected outside of the human body, on the human body and within the human body.

The institutionalization of mass surveillance as an important mean for control, detection and order has become common use in modern societies to carry on sovereign discourse. For instance, Echelon, the first known comprehensive mass surveillance program of the communication age that was established in co-operation of the United States, Britain, Canada, Australia and New Zealand, during the Cold War was used for military and diplomatic purposes in the early 20th century. But its use was expanded in late 20th century to listen to private and commercial communications infrastructures [17]. On the other hand, as it was disclosed in 2013, most U.S.-based global digital service companies' databases have been accessed within the scope of mass surveillance practices conducted through a program called PRISM [18]. Amid growing concerns about mass and unauthorized digital surveillance, the UN General Assembly's 68th session of the 2013 meeting held in 2013. A/RES/68/167 Decision on privacy in the digital age was taken in the session [19].

With the interaction of organic elements of real people with cyberspace, the organic integrity of traditionally accepted human beings is taking a new form. As progress has taken place in related areas such as information technology and biotechnology, human integrity is gradually evolving into cyber-organic forms.

Following the development of digital data-based computing, automated information processing-based activities have brought up artificial intelligence discussions. Currently, the legal personality of artificially intelligent systems is out of the question. However, discussions in this direction continue. For instance, The European Parliament's Decision of 17 February 2017 has taken an important step in how autonomous robots will be represented in social and economic activities and what kind of responsibilities they will carry. This Resolution recommends legal status under the definition of "electronic personhood" for autonomous systems [20]. On the other hand, In 2017, for the first time in history, a robot named "Sophia", with a partial artificial intelligence, was granted citizenship by the Saudi Arabian government [21], although there is no social acceptance in real.

C. Transformation of Activities

Each activity includes a population unit and part of space. However, in cases where the activity has factual properties, it is possible to make an assessment independent of the environment and the actor. Modern society is based on the differentiation of social roles. It is also possible to consider all social activities within the scope of political, economic and cultural activities.

The expanding social formations with globalization and the development of transportation opportunities have caused real and legal persons to move and interact more and more within the different sovereign areas. As a result of digital technologies, people have become able to engage in social interactions without going from one place to another physically. Unlike other technological advances, digitization leads to new developments that significantly affect both organizations and human preferences at the same time. Within this perspective, a very comprehensive sphere of influence has emerged for states.

The uncertainties and change surrounding the domestic or foreign status of the active actors disperse the relationships between these actors through network structures while at the

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same time transforming the distinct line that exists in terms of sovereignty [22].

III. Towards A Conceptual Framework For Digital Data Sovereignty

With the digital data sovereignty, the authority of the state on the digital data produced, processed and stored in the field of sovereignty is not discussed. The increasingly visible debate is how to rebuild the state's eroded control practices on the growing volume of digital data and accelerated data traffic.

Digital data sovereignty includes discussions about which state sovereignty and therefore the laws governing the digital data generated. The concept of digital data sovereignty is evolving against security risks, especially for both space and society. However, there is no accepted common definition yet [23]. Meanwhile, the digital data sovereignty debate is focused on the acceptance that digital data will be stored and processed within the boundaries of a specific country physically [24] and the state's authority over information assets [25].

One of the most important topics of discussion in recent years, "protection of personal data" and "digital data privacy" is an individual-level debate, while increasingly gaining political and national character to the rhetoric of "national security" and "digital data sovereignty" has begun to evolve. According to Polcak and Svantesson [3], although they belong to different legal and political areas, there is a close relationship between digital data privacy and digital data sovereignty. While digital data privacy revolves around individual rights, digital data sovereignty comes into the context of state authority. However, it is possible to evaluate digital data privacy in the context of individual sovereignty. From this point of view, with "digital data sovereignty", states are trying to define the external non-intervention area of sovereignty.

Although the main course of the digital data sovereignty debate is on the axis of national security, there are also controversies that have developed in response to the gradual erosion of collection, ownership and applications as the sovereign rights of indigenous peoples (such as Indian tribes in the United States, Aborigines in Australia) over their territory and people, [26].

Technically, digital data has no nationality and is subject to the legal requirements applicable to the physical space in which it is stored [27]. However, as the perception of risk generated and processed in the field of sovereignty of a country increases, efforts to nationalize and put digital data under authority in discourse and practice.

The issue of digital data sovereignty has been increasingly discussed and taken globally. The reasons for the increasing debate are as follows:

- A proliferation of examples of digital data of social activities in a sphere of sovereignty that can be interfered with by other governments and service providers, and this supports the perception of threats to national security.
- Advances in next-generation information technologies rapidly increase the volume of digital data and complicate the control of digital data, increasing the possibility and likelihood of intervention
- Increased centralization in platforms where social and economic activities based on cyberspace are carried out and this centralization brings specific companies to the forefront and relations between companies and governments create new dependencies.

Information technology resources of end-user individuals, private companies and public institutions are increasingly being transported towards centralized systems with enormous computing power and storage capacity due to cost advantage and flexibility. However, users' control capacity over infrastructures and digital data is also decreasing. As the number of users who do not understand and manage information technology infrastructures increases, these technologies come under the control of those who own infrastructures and know how to manage them [27].

A. Sovereignty and Information Systems

The ability to manage information is known to have an important role in the formation and transformation of modern states by strengthening authority structures [28]. The claim of sovereignty of states and the validity of the claim of sovereignty requires knowledge as a priority. In this context, it is problematic for the states to claim sovereignty over things that they do not technically know, and to implement the necessary practices.

The relationship between sovereignty and information technology in literature, cyberspace sovereignty [29], digital data sovereignty [24], information sovereignty [30] control [31], mass surveillance [32], crossborder information flows with Internet [33], governance of global networks [34] and answers were sought under different headings. However, the current literature is quite dispersed and does not give a holistic perspective because it brings interpretations through an application area after discussing the concept of sovereignty in general.

Modern state crises transformed by information technologies and globalization have been contacted in the literature in different topics. For example, the state's changing position in the context of cyber-security in the information age has become an important topic in this sense [35]. According to Singh [36], there are two main influences on the sovereignty of information and communication technologies:

- ICT and the state's ability to dominate existing spaces and population swelled.
- The cyberspace environment shaped by ICT has created a new place of domination for states.

The collection of large data generated by the digitization process on a transnational scale obscures definition of what is national, thereby blurring the boundaries between security and intelligence. Accordingly, there is a transition from small amounts of digital data to large amounts of digital data with high uncertainty by drinking high precision in the fight against crime [22].

B. Cyberspace versus Digital Data Sovereignty

The starting point of the cyberspace sovereignty debate sparks the desire of states to have sovereignty over all the places where social activities take place. It is understood through the size of the space of sovereignty and reflected in the practices of the state. Since there is a state-centric perspective, there is no equivalent to discussions in this

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direction at the individual level in general. There has been a growing rhetoric against the possibility of outside intervention by other countries. Cyberspace sovereignty is technically a broad topic because it covers all the infrastructures that make cyberspace possible. On the other hand, digital data sovereignty, with the increase in data-based social activities, and the protection of personal data and privacy through individual-level discussions and state secrets on behalf of the state it became a concept against outside interventions and began to reflect on state practices. Digital data can generally be considered as a subset of cyberspace domination [24]. Indeed, it generally overlaps with the technical infrastructure of cyberspace. However, it is not possible to conceptually limit the sovereignty of digital data to cyberspace space. Digital data stored in physical spaces and isolated from any communication network is also conceptually covered by digital data sovereignty.

C. State Practices

Policies for localizing digital data traffic, which is a state secret, have been on the agenda of all countries since the earliest periods of the Internet. However, countries such as Russia, China, Iran, North Korea, especially in the case of national security in case of interference and the superiority of U.S.-based companies in information technologies, more aggressive policies for the realization of within the territory of sovereignty. However, in 2013, the u.S. revelations that it collected and analyzed data from other states for the purpose of intelligence and industrial espionage against terrorism, European countries such as Germany and more democratically positioned countries also take more concrete steps in digital data sovereignty [5, 24, 37]. States also go to regulations in "authority technologies" in response to sovereign rights and security problems. Below are the practices of different countries to ensure digital data sovereignty:

- Regulations applicable to the Personal Data Act in Force in Russia since 2006, foreign or domestic companies are required to store and process personal data belonging to Russian citizens within Russian borders is being made available. It has evolved as a measure against increased sovereignty over personal data and national security risks [38].
- Responding to the threats, China may face in the use and exploration of deep waters, space, cyberspace and polar regions with its assets and activities in the country's borders and areas in the National Security Act revised in 2015 authority is defined. At the same time, the Cyber Security Act, which came into force on July 1, 2017, introduces detailed regulations on the protection of sovereignty in cyberspace and sanctions [39].
- The General Data Protection Regulation (GDPR), which entered into force on 25 May 2018, has ruled that digital data of EU citizens must remain within EU borders [40].

Similar legislative regulations on digital data sovereignty and localization efforts are also available in Anglo-Saxon countries such as Australia [37]. Countries naturally have different policies and legal regimens on digital data. However, this situation leads states even with good relations to be more cautious towards each other. In 2007, the French administration ensured that senior executives were noticed that email data was encrypted and advising them not to use a Blackberry phone in allied countries such as the United States, Britain and Canada [41].

Maintaining digital data sovereignty is not as easy as desired by the technical facilities available on open networks (such as the Internet) that are not under control. The physical location of the server in which digital data is kept must prove that digital data is stored on this server [25]. The proposed technical solutions for digital data sovereignty are briefly listed below [5, 42]:

- National e-mail service: Conducting local e-mail services and all e-mail correspondence through servers within the country
- New transit cables: New transit fiber lines between countries, regions and continents
- Internet routing: Conducting and directing domestic data flow in a geographically restricted area
- National cloud storage: Implementing private and public solutions for cloud storage services
- Encryption tools: Better encryption of end-to-end communications

IV. Conclusion

It has been tried to be found that ensuring sovereignty is either a social argument that can be realized with purely political discourse or a technical issue that is easy to determine with expertise in digital technologies. Sovereignty, in the broadest sense, gives power the authority to control and determine the outcome of everything within the field of sovereignty. Naturally, each technological development brings different possibilities and possibilities. This general equation also applies to the impact of digital data technologies on the sovereignty of the world. Since the subject of discussion of the study is the changing areas of sovereignty on an international scale and the reactions given by the nation state to this, the existence of discussions on the transformation of the concept of sovereignty is noted and the subject is left to another study. The concept of sovereignty requires mutual recognition between the state level, or the actor concerned with the state. The claim of sovereignty of states and the validity of the claim of sovereignty requires knowing as a priority. The state cannot claim sovereignty over things it does not know. Digital data sovereignty has become emerging topic with respect to socio-economic developments. But it is much more difficult than technically positioning a country within cyber-walls. In following studies, technical and political essence of digital data technologies will be scrutinized in order to have a better understanding of changing nature of the concept of state sovereignty.

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